THE UNIVERSITY OF
SYDNEY

Friendship On Wheels

ISYS2110 - Group 3

SID:
480405724
520434673
530528603
510574673
530078566

Executive Summary

Friendship on Wheels is a service looking to provide meals and care to those who need it most in our community.

As the essence of our company, we believe in fostering a healthy and thriving community and hence our core values are:

- Prioritising the mental and physical well-being of our community members through nutritious meal delivery and social connections
- A platform to provide genuine care and empathy, fostering a supportive and compassionate environment for everyone we serve
- Building strong, connected communities by encouraging volunteerism and ensuring that everyone feels valued and supported

The system requires the requester to be able to order meals, and request relevant services. volunteers to check and deliver the orders and fulfil requests for extra services, and administrators to assign volunteers to a requester (for no longer than 1 year due to security and safety reasons) and run general maintenance and facilitate the website.
Users will access the system through our web-based application, which includes the aforementioned ordering and volunteering systems.

Our analysis of our target demographic indicates there are two volunteers for each requester, which would result in greater flexibility for requesters or admins to choose a volunteer.
Also, as the survey showed a significant demand for a feedback form, we have incorporated it into our system with other standard features such as company introductory pages and frequently asked questions pages.

Our web interface design is intended to be usable for everybody, especially our older users, who make up a majority of our prospective requester user base. The use of large fonts, high-contrast background colours and easy-to-navigate pages ensure that anybody of any technological literacy is able to use our site.

It is the basis of helping others in need which drives us to create and provide Friendship on Wheels to people in need who are looking for a meal and a friendly visit. Our system aims to fill a spot of need for the people in our society who are most at risk.

Scope:

Friendship on Wheels (Friendship-On-Wheels) is our web-based system which aims to connect requesters and volunteers. Specific definitions of requesters and volunteers will be covered under the user demographics.

Friendship on Wheels will provide multiple services from Volunteers to Requesters, which these services will include:

- Meal delivery
- Transportation Services
- Light housekeeping services
- Companionship

With Meal delivery as our main service, all additional services are to be requested as an “add on" service to the meal order. Requesters will be able to place additional services on top of their meal orders only, to which services other than meal order are not available on their own.

Through the exploration of the website, all users will be able to discover and learn the services provided by Friendship-On-Wheels and get involved through any measures. The website will be built in consideration of all users and address each requirement of users through the design and functionalities. To cater to older user demographic and users with visual impairment, the website will be of simple yet concise layout with vivid colours and large texts to provide clear directions to navigate through the website. The design highly focuses on a user-friendly interface, supported by functional requirements to ensure flawless interactions for all users on their needs.

The website will also be managed by administrators to overview the user usages, maintain and overlook user signups, operations and scheduling of each user, and lastly, oversee the requests and adjustments for requester orders. The administrators will overlook the entire scheduling system for both requester and volunteer and make sure that there are no errors or fall outs in service provision with no overlapping schedules.

Highlighting the security and safety of all users, we aim to security check all users and their backgrounds to provide a safe working environment for all volunteers, and support a comfortable experience for all requesters. Supporting a secure environment, all volunteering hours and activities for each volunteer will be kept on record for both security and data collection purposes. This data also includes meal requests and other service requests from requesters, where each user can access the record of their service request or volunteer hours themselves through the website.
To ensure the safety of all users and prepare for any unexpected, urgent situations, we aim to provide volunteers with access to programs for essential medical training for first aid. This will be pursued by connecting volunteers with organisations that will help them through the process of obtaining various medical certifications and training.

Based on the requester's preferences and requirements, Friendship-On-Wheels will assist in assigning volunteers with necessary training and certifications to the requesters. As mentioned previously, these volunteers will undergo multiple background and security checks, which will be verified by our third-party "ConnectID" to ensure the safety of the requester and vice versa.

The process of assignment will continue until both parties are happy with the pair that has been formed. Friendship-On-Wheels pledges that at no time during the selection process will the requesters have access to the volunteer's information. We will maintain the privacy and confidentiality of all parties involved in the process and corresponding information will only be available after a match has been successfully made. After the pair has been made, the volunteer and requester will only be able to access required information to execute service deliveries. Other critical information will be kept private and will not be open to others unless there deems to be a critical emergency in requirement of access to such information.

Both requester and volunteer will be able to log their availability in the website once they have signed into their accounts. Requester and Volunteer website functionalities will be differentiated to perform the requirements of each corresponding party. Requesters will be able to request services through the website, whereas Volunteers will be able to schedule and log their volunteer hours instead.

The Friendship-On-Wheels website will assist all users and any stakeholders to easily navigate through the website and understand the services and operations provided by Friendship-On-Wheels. The website will provide requester/volunteer registration and login to the website, and upon login, requesters and volunteers will be able to access different functionalities suited to their needs.

User Demographics:

Our system will majorly tailor for three key groups: requesters, administrators and volunteers. The age range and the ratio between volunteers and requesters have been identified through the results of our initial survey, where it shows a 2:1 ratio between the volunteers and the requester. Further details of survey results will be discussed in the analysis section.

Requesters can be categorised as the elderly, disabled or individuals with special needs in requirement of medical services, transportation services, companionship, or logistical support. The requesters are classified as the actors that receive the service from our varieties of offered service selections. The requester will be the primary user of the website where all services can be requested through their personal accounts.

The volunteers will be any capable personnel that have passed the security check to support in areas which are most suited to the characteristics and abilities of the volunteer. Volunteers will be classified as the actors that provided the services to requesters, in multiple forms of delivery services, transportation services, personal assistance to companionship. The Volunteers will also be another primary user of the website where their scheduling and work time can be managed through their personal accounts.

Administrators will be staff equipped with comprehensive training and knowledge for efficient system management. The Administrators' roles will include general management of the website, management of all Requester and Volunteer profiles and schedules, service offerings and data recording. The administrator will ensure and implement security checks and seamless operation across all platforms.

We will be addressing each specific needs of each group to ensure the system is user-friendly, secure and adaptable to have an enhanced service delivery.

Functional & Non-Functional Requirements

Here is a list of all our functional requirements including fundamental features such as user registration, authentication and login, and further notable functionalities such as meal ordering and volunteer availability management. Additionally, we included supplementary elements such as the option to cancel, provide feedback and access answers to FAQs to enhance user experience and support.

Functional Requirements

1. The system **must** facilitate user registration, allowing individuals to create accounts as either requester or volunteers.
2. Upon account creation, users **must** have seamless access to sign in and out of their respective accounts.
3. The system **must** allow users to update their personal information as necessary
4. The system **must** display contact information to allow users to reach out to our friendship on wheels team.
5. The system **must** display a comprehensive FAQ page addressing commonly asked questions, supplemented with resources for additional support.
6. The system **must** provide an option for an online feedback submission form, allowing users to provide comments.
7. The system **must** include a mechanism which allows volunteers to log their availability and view assigned tasks on a timesheet.
8. The system **must** incorporate a feature which allows users to cancel their appointments within a 24hr window prior to the scheduled time.
9. The system **must** allow requesters to order meals / visits / safety checks.
10. The system **should** offer a live chat feature to provide immediate assistance to users while using the website.
11. The system **should** allow volunteers to upload any documents onto their profile → medical certificate etc

12. The system **could** include a live tracking map feature → to identify requested/volunteers near each other to pick from → order confirmation + live tracking will begin once volunteer starts the journey
13. The system **could** maintain a record of all previous requests made by requesters, storing relevant details in a JSON file format for efficient handling and storage.
14. The system **could** incorporate multi-language support, allowing users to select their preferred language from a dropdown menu and enabling real-time translation for seamless communication.
15. The system **could** include an intelligent meal recommendation engine that analyses requesters' dietary preferences, restrictions, and nutritional needs to suggest personalised meal options.

Non-Functional Requirements

Our non-functional requirements require our website to be capable of handling a high volume of users and requests without compromising performance and resilience while also accommodating for future scalability and improvements. As we have a diverse user-base, we need to ensure that our website is intuitive, user-friendly and easy to navigate for all. Additionally, we aim to adhere to current security protocols for user privacy by limiting user accessibility to other user data and implementing data backup, hashing users' details, and other cryptographic methods.

Here are 6 of our non-functional requirements listed below.

1. The system **should** be capable of handling a large volume of users and requests without significant degradation in performance.

   In preparation for instances of overloaded website use, the system should be capable of handling the large volume of users without having its performance largely impacted.

2. The system **should** have a high level of reliability to ensure continuous availability and minimise downtime.

   As a general requirement, the system should be able to continuously produce the correct results, without encountering errors or crashing. The system should be able to handle all errors and with constant maintenance and upkeep, it should be able to provide definite reliability for use and minimise any downtimes.

3. The system **must** be intuitive, user-friendly and easily navigable to cater to the diverse needs and technical abilities of all types of users (requester, volunteers, administrators).

   For all users, the system must be intuitive and user-friendly. It should be easily comprehensible and usable. This non-functional requirement will be implemented throughout the website with the choice of colours, fonts, use of images and simple layout.

4. The system **should** implement high-security measures to protect users’ details.

    As another general standard, the system should provide high security measures to ensure all private data of each user is properly encrypted and protected. The system will ensure to have no data leakage and promise safety from data hacking.

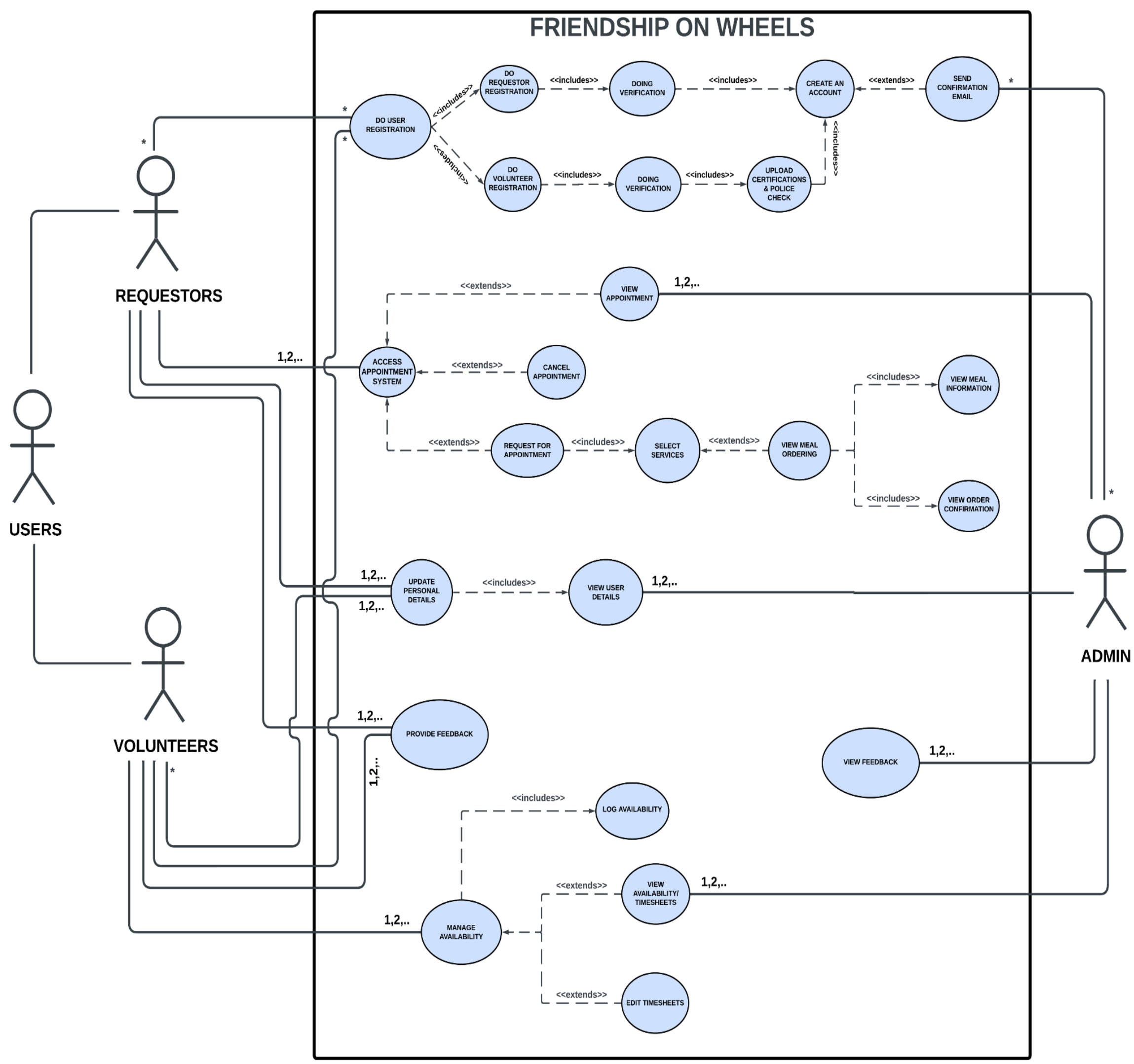
5. The system **should** be designed to accommodate for future scalability and improvements.

    Similarly to 1, for future expansion and increase in users, the system should be able to instantly adapt and adjust itself to accommodate the increased usage.

6. The system **should** implement measures for data backup and disaster recovery to ensure quick restoration of service in the event of a system failure.

    In case of data loss, system failure and accidental deletion of accounts on the users’ side, the system should be able to access data that has been continuously backed up. All vital data should be properly backed up regularly to avoid imminent loss and will be maintained with high security to guarantee no data leakage.

Use Case Diagram



~ [link to use-case diagram]

The use-case diagram is made to capture the overall user interactions with our friendship on wheels website (system). This helped us in mapping out the features which are required by the system, hence, fulfilling the system's functionalities. The diagram further helped in better understanding the functional and non-functional requirements of the system then deciding which features are deemed of utmost importance for the users, which then was included in our website for use. During the interim presentation, we've received feedback to add multiplicity labelled verb(s) and noun(s) to the use cases. We positively took this feedback and incorporated these changes to improve the existing use-case diagram accordingly. Multiplicity has been added to the main use-cases which interact directly to the users and the admin. The proper use of verb(s) and noun(s) for our labelling gives us a clear insight of each use-case activity.

Our website aims to carry out the following main tasks:

USE-CASE 1:

- Users (which consists of volunteers and requesters) should be able to register themselves and the administration should send a confirmation email once their profile/account has been created. Given the fact that we have a diverse user-base spanning from 25 to 80 years on average - it becomes necessary for us to ensure that the registration process is not only simple with few steps but also ensure seamless logins and log-outs from their accounts. [FR 1, 2] [Nielsen's laws 1, 2, 3, 4]

USE-CASE 2:

- Requesters can schedule an appointment by selecting their preferred service (which includes safety checks, friendly visits and meal delivery) after which they can fill in their meal preference. Additionally, they can view the appointment details (such as date, time and name of the volunteer) along with the meal information on the website. They will also have the option to cancel their appointment 24-hrs before the scheduled time. [FR 8, 9] [Nielsen's laws 1, 2, 3, 4, 6]

USE-CASE 3:

- Volunteers can also easily manage their availability by completing their timesheets. They also have the convenient option to make edits to their shifts (scheduled volunteer hours) at any time, which can be accessed from their personal profile or account until 24-hrs before the service-time. This gives them the necessary flexibility to change their schedule if any unforeseen circumstances arise. Furthermore, it allows the administration to ensure that there aren't any sudden service cancellations before the scheduled service. If there are instances of absolute last minute emergency for cancellation, the volunteers and the requesters will be suggested to get in touch with the admin to organise the following steps after cancellation. [FR 7, 8] [Nielsen's laws 1, 2, 3, 4, 6]

Other tasks also includes:

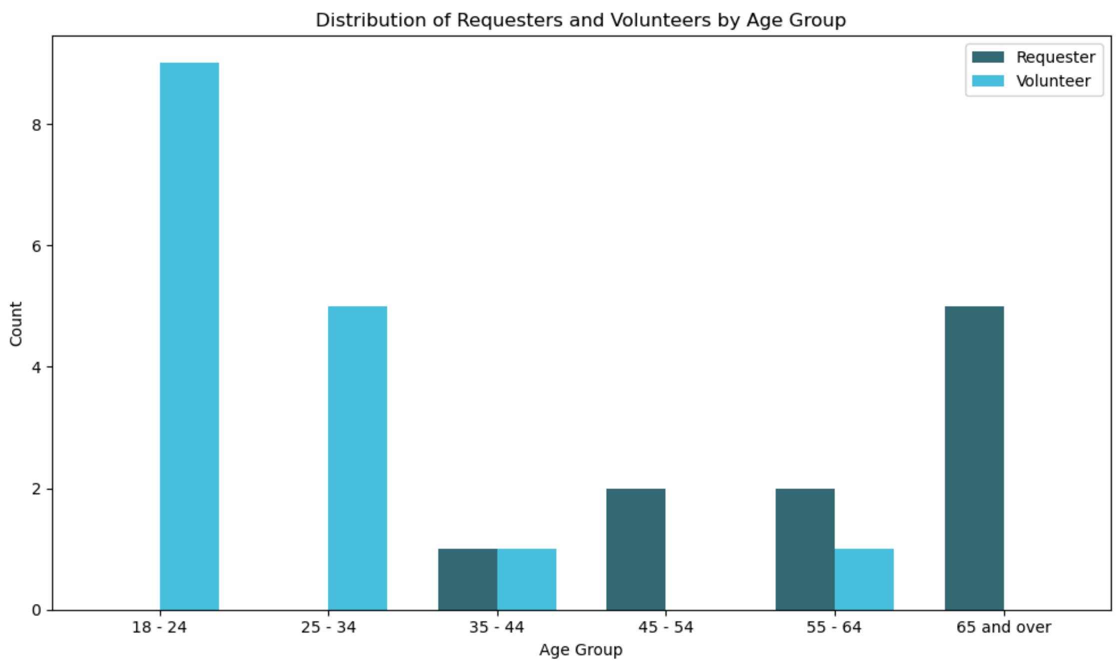
- Registered users should be able to update their personal details. This includes essential information like their name (first and last name), date of birth, email, gender, address, accessibility issues, preferred language as well as their preferred volunteer and meal preferences. [FR 3] [Nielsen's laws 1, 2, 3]
- Users are able to provide their feedback on the services they partook in. Additionally, 'FAQ', 'Our Services', 'Contact Us' and 'About Us' pages have been created so as to improve customer experience for all the users who are visiting our website and/or are registering for any of the services being offered by us. [FR 4, 5, 6] [Nielsen's laws 1, 2, 4, 5]

Our two actors of Requesters and Volunteers are then matched based on their availability and their service preferences. It is also kept in mind that the matching preferences of the requester

are taken into consideration while matching them with a volunteer and the Volunteers' preferences and opinions are also clearly respected.
All these consideration processes centralised on the comfort of each actor helped us to draw the functional and non-functional requirements for this project and guide us in the development process by looking at the interactions and primary functionalities required by the system in the early phase of the design.

Analysis of Survey Findings

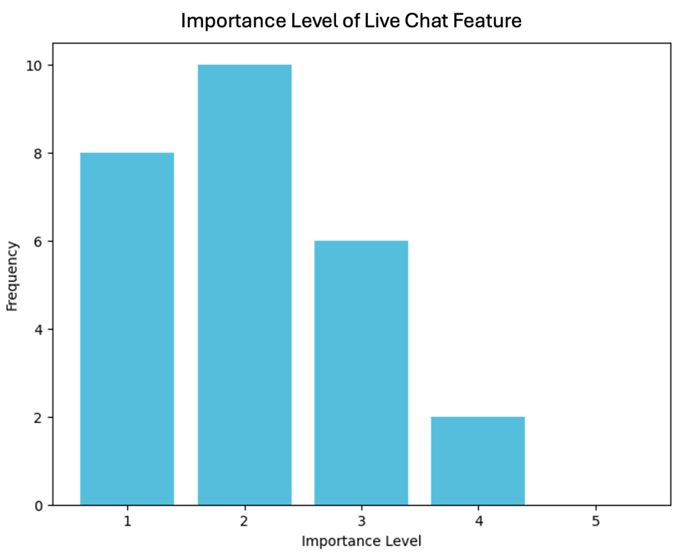
To complement our user-friendly interface and better understand our audience, a survey was taken among 27 individuals to incorporate the findings into our web-app requirements.



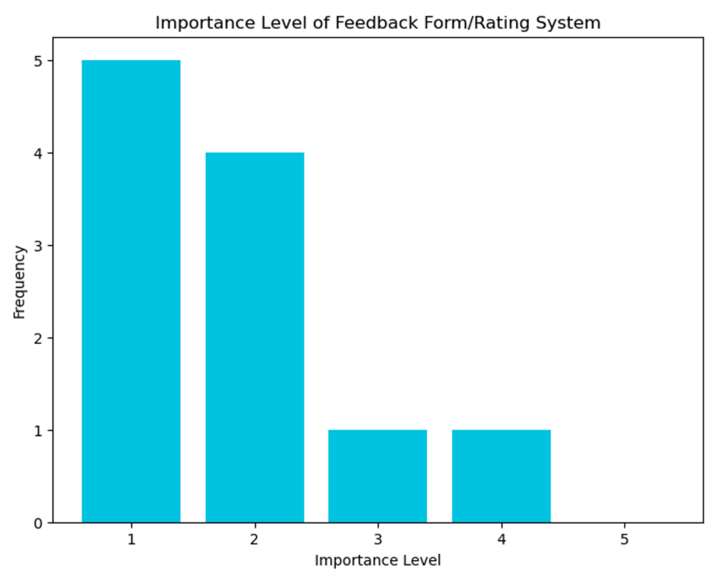
As evident from the result shown above, the requester to volunteer ratio is almost 1:2, meaning that for every requester using our platform, there can be at least 2 volunteers allocated to provide assistance to each requester. This is the precise balance we aim to achieve in our system as it accommodates our goal to create a platform that empowers volunteers but also ensures that requesters are provided the support they need, when they need it, in any circumstances.

This insight further allows us to prioritise features tailored to the needs of both types of users. We aim to equip volunteers with a mechanism to view and accept requests, as well as adding their availability to a timesheet, making it easier for administrators to pair up requesters and volunteers based on their availability. Hence, this volunteer to requester ratio is a crucial form of data to ensure we can provide flexibility to both parties. Ensuring that there are a sufficient number of volunteers for each requester can promise features of convenient rescheduling and cancellation of service appointments. This rescheduling and cancellation feature is beneficial to both parties in promising that in any situations of emergencies and/or misfortunate events, there are always stand-by volunteers to cover appointed services without disrupting everyones' schedule.

Our analysis also reveals that the majority of the volunteers are young adults aged 18-35 whilst all our requesters fall in the age bracket of 35 and above. This emphasises the importance of tailoring our platform towards the older demographic, in line with our NFR 3, by incorporating larger fonts, clear navigation menus, live chat feature and intuitive interfaces for easier accessibility. This adoption can be further supported by a study conducted by the PRC stating that only 32% of adults aged 65 and older report feeling confident when using electronic devices, compared to 87% of young adults.



← Survey result collected from all survey participants

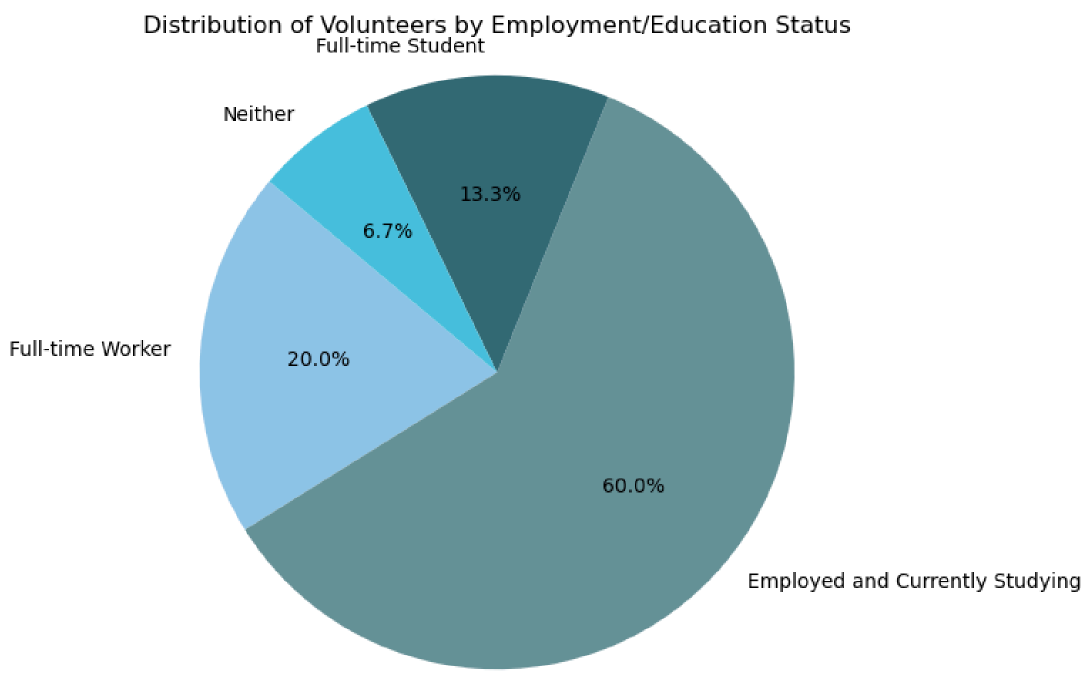


Survey results collected from Requestors →

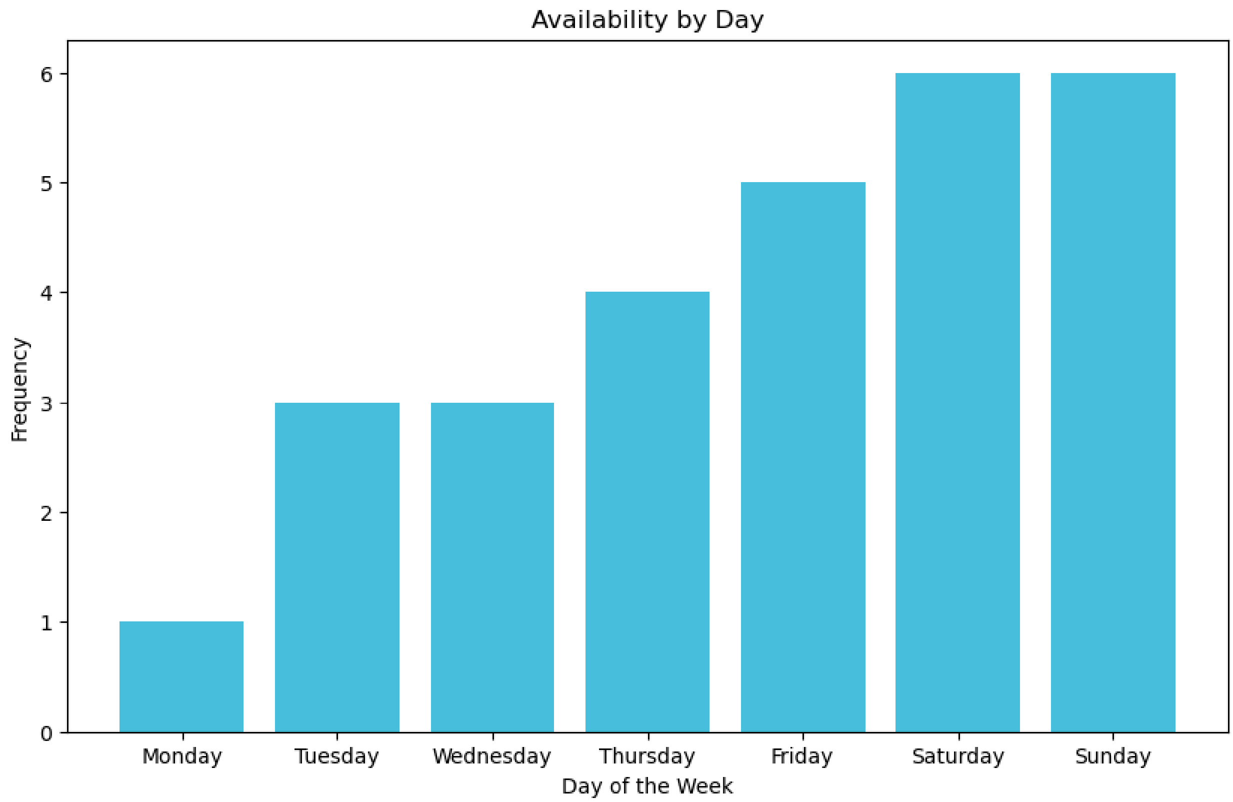
As the graph above illustrates, on a scale of 1 (being most important) to 5 (unimportant), the data is skewed towards the left, showcasing that respondents would greatly benefit from having live chat assistance in usage of our website. With incorporation of a live chat feature and to further support the requirements strongly expressed by the majority, we've added a FAQ page which includes commonly asked questions as well as additional resources for further guidance, aligning with the 5th and 9th FR. These frequently asked questions have been pooled from basic questions that may arise in regards to the use of the website and services provided by the business. Our FAQ page will provide quicker and faster access to the answers of common questions, whereas our live chat feature will assist users in achieving answers required for their specific questions and needs.

Moreover, we also asked our requester respondents how important it is for them to be able to provide feedback on the service or the volunteer they were matched with. Again, we've received a favourable response where the majority would like to have such a system, which conforms with our 6th FR. Hence, we are committed to incorporating this feature to ensure transparency, accountability, and continual improvement of our service.

In terms of making our web-app friendly and efficient towards volunteers, we've decided to include a timesheet system, scheduling calendar as well as extra resources to gain medical training, upon request. The way we envision this to work is that the volunteers would notify us of their interest in receiving medical training, and then our admin staff will connect them to a third-party health training company we are allied with. This partnership would ensure that volunteers receive high-quality, accredited medical training.



The analysis of our volunteer demographics reveals a prevalent trend among participants, with many balancing multiple commitments such as employment and full-time university alongside their volunteer work. This underscores the significance of offering flexible volunteer opportunities and scheduling to accommodate the diverse lifestyles of our volunteers, ensuring their continued engagement and participation.

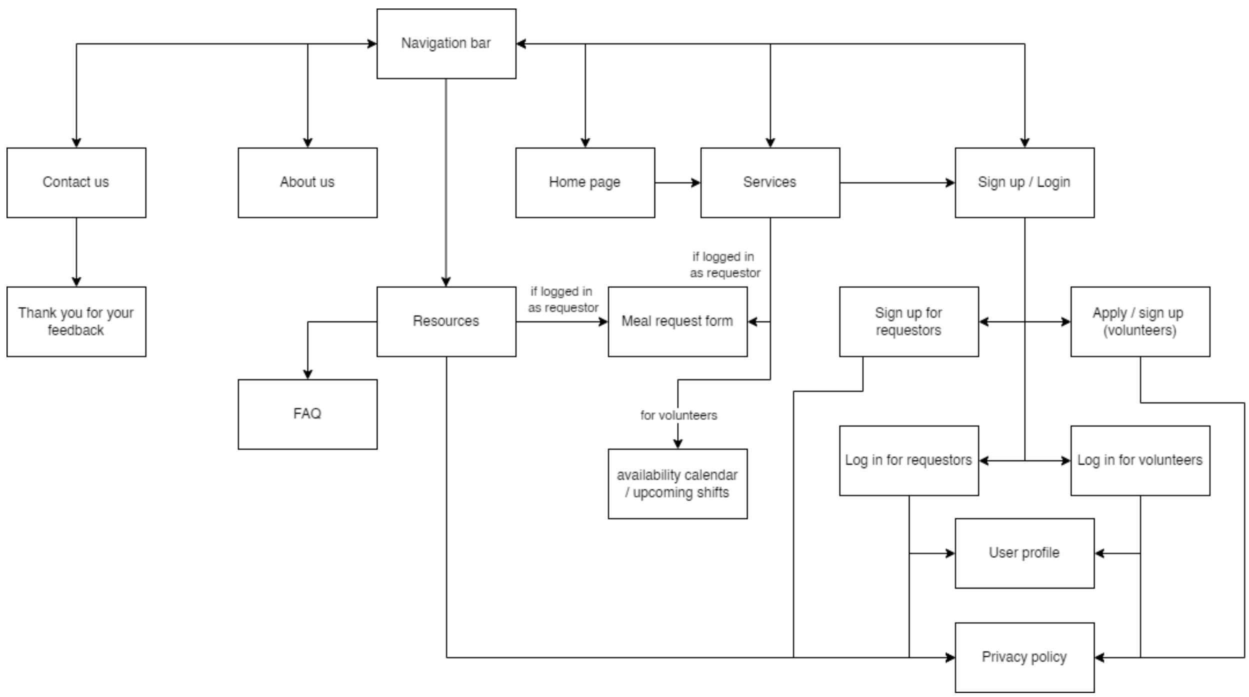


Furthermore, a deeper examination of volunteers' availability (in above graph) highlights a notable concentration of availability on weekends and Fridays. This finding underscores the importance of implementing a timesheet system, to streamline scheduling processes and optimise volunteer participation. By effectively aligning volunteer availability with requester needs, we can enhance service delivery and ensure timely support for our community members. Thus, this insight serves as a catalyst for the inclusion of our 7th Functional Requirement, aimed at enhancing scheduling efficiency and maximising volunteer engagement.

Design

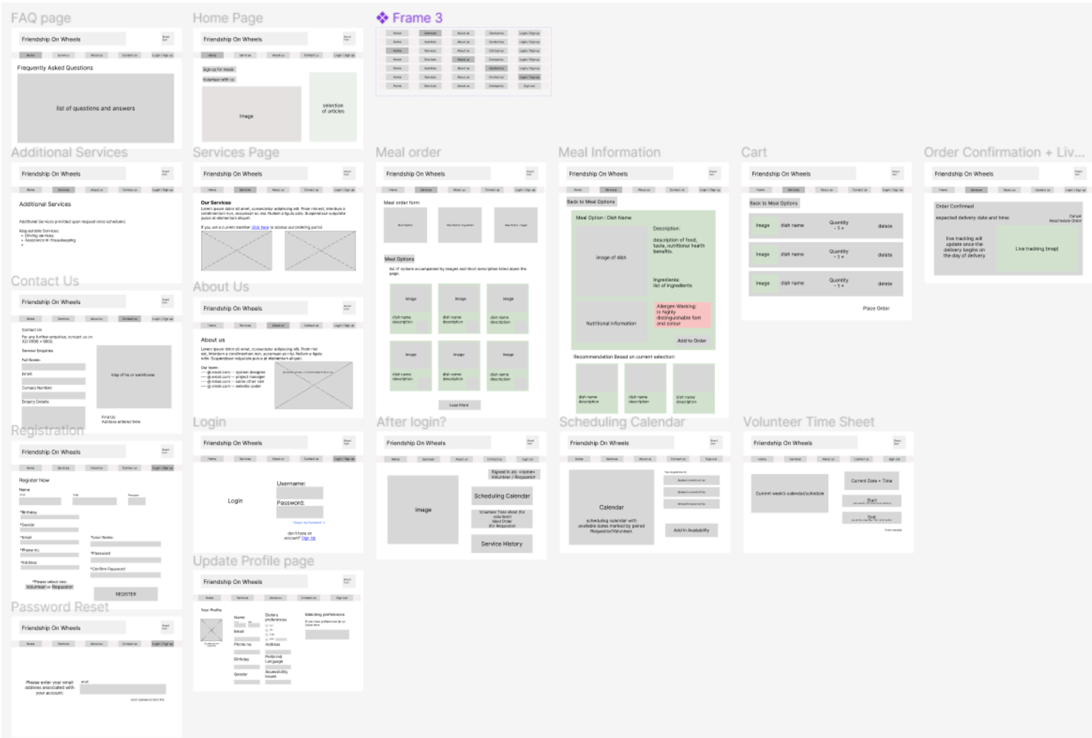
Site Map

Our site map outlines the general connections between the functions of our site. As our aim is to provide easily navigable and accessible service pages, the meal request form had been planned to be accessible from both the resources and services page. However, in our final design, the resources page has been excluded, and the meal order and timesheet has been made available from the user profile instead. This decision was made to ensure that only the correct corresponding user will be able to access their required pages and reduce any confusion and/or misuse.



The page hierarchy can be clearly seen from the sitemap where a lot of pages can be accessed from the “Account” page and “Services” page. Once the Sign Up/Login page is accessed, a separate Sign Up page can be accessed to create an account for users without an account. Then, upon logging into the users’ personal accounts, users can access their profile and different services pages depending on their account type. Service pages vary for each actor (volunteers and requesters), where requesters will be able to access meal order pages with cart, and volunteers will be able to access timesheets to log in their working hours. These pages will only be available to each corresponding account type and scheduling calendar and volunteer timesheets will also only be accessible after logging in.
All other relevant pages of “Contact Us”, “About Us” and general information can be found and accessed from the initial Homepage.

Initial design - wireframe



~ [link to wireframe]

A link to the online version of the wireframe can be found in the appendix. This is our initial design of the website, reflecting the integration of our survey findings which indicate a strong preference for PC usage in accessing the internet.

The initial wireframe began with a couple selections of web pages as our initial prototype design. As we confirmed our services and functionalities of the website, we added wireframes of our additional pages to serve as a guideline when building our actual website.

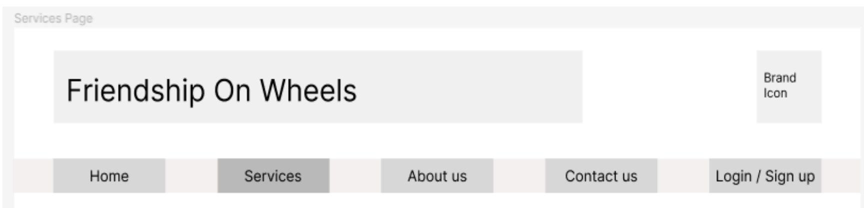
The design of the website primarily focused on the usability for all users, where simple layout with precise directories and information was the key to our website design to have a friendly interface. The initial wireframe design greatly focused on implementing the functional and non-functional requirements. Basing off these initial prototype designs, we've built our website accordingly with additional features of enlarged texts and varying use of colours.

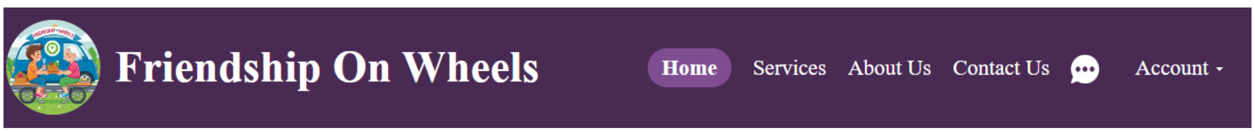
Usability

Our website has been meticulously designed to prioritise user experience, catered with the gathered data from our survey findings. The webpages feature a simple yet concise layout with clear, large fonts tailored to ensure readability for any user, with particular attention to the needs of our older users. To improve the website usability and for easier understanding, images of services and products have been incorporated throughout the website with corresponding descriptions.

In addition, interactive elements such as buttons and directories (with hyperlinks between articles seamlessly guiding users to relevant content) are strategically placed with clear size and colour distinctions. The choice of high-contrasted purple colour scheme enhances visual clarity, adhering to the HCI principles for improved accessibility and engagement. These design decisions collectively contribute to an intuitive interface, meeting our third non-functional requirement of delivering a user-friendly browsing experience.

In adapting our initial wireframe to our final design, we implemented specific changes to the layout and style of the webpages to enhance user experience. Specifically, our web system's header serves as the first thing users see across all our web pages, therefore we wanted to ensure it was both visually appealing and efficient in navigation.





With this goal in mind, we deliberately changed the positioning of the components within the header, such as relocating the logo and website title to the left side. This strategic move provides us with an intuitive visual hierarchy, but also frees up valuable space for content below.

Furthermore, we consolidated the main page links into a single line format which is in-line with the logo and web title to optimise accessibility while maintaining a clean and uncluttered appearance. This simplification enhances user comprehension and navigation efficiency, which is a core aspect of creating an effective user interface *(Babich, N. 2016)* that aligns with the principle of making interactions clear and concise, ultimately leading to a more intuitive and user-friendly design. These links are highlighted to allow users to visually see which page they are currently on. For example, as seen above, when a user is on the homepage, the ‘Home’ button will be highlighted, providing immediate visual feedback. This feature not only helps users navigate more easily, but also enhances their overall experience by reducing confusion and ensuring they always know their location within the site.

Our header features a contrasting yet pleasant purple background colour, allowing for users to easily read the text and links displayed. This consistent design choice, adhering to HCI principles, ensures the simplest navigation experience across all pages, as users encounter the same familiar header layout and style throughout the website. Users can access any page within three clicks directly from the header *(Poon, J. 2024)*, minimising the effort required to find desired content and further supporting Nielsen’s rule of recognition rather than recall, as it provides easy accessibility to a wide variety of pages.

About Us Page

Our ‘About Us’ page gives users a chance to understand the team behind Friendship on Wheels, which provides a sense of trust, establishing the connection between the user and our service. Acting as a first impression, this page plays a crucial role in shaping user perceptions and building rapport.

Contact Us Page

In alignment with our user-centred approach, the ‘Contact Us’ page offers users a seamless avenue to provide feedback or make general inquiries about our services. This feature is a direct response to user feedback, ensuring that our platform remains responsive to user needs. Additionally, users have the option to access our office location through an integrated map, catering to those who prefer in-person discussions. This holistic approach to user interaction reflects our commitment to delivering an accessible and streamlined interaction.

User Specific Web Access

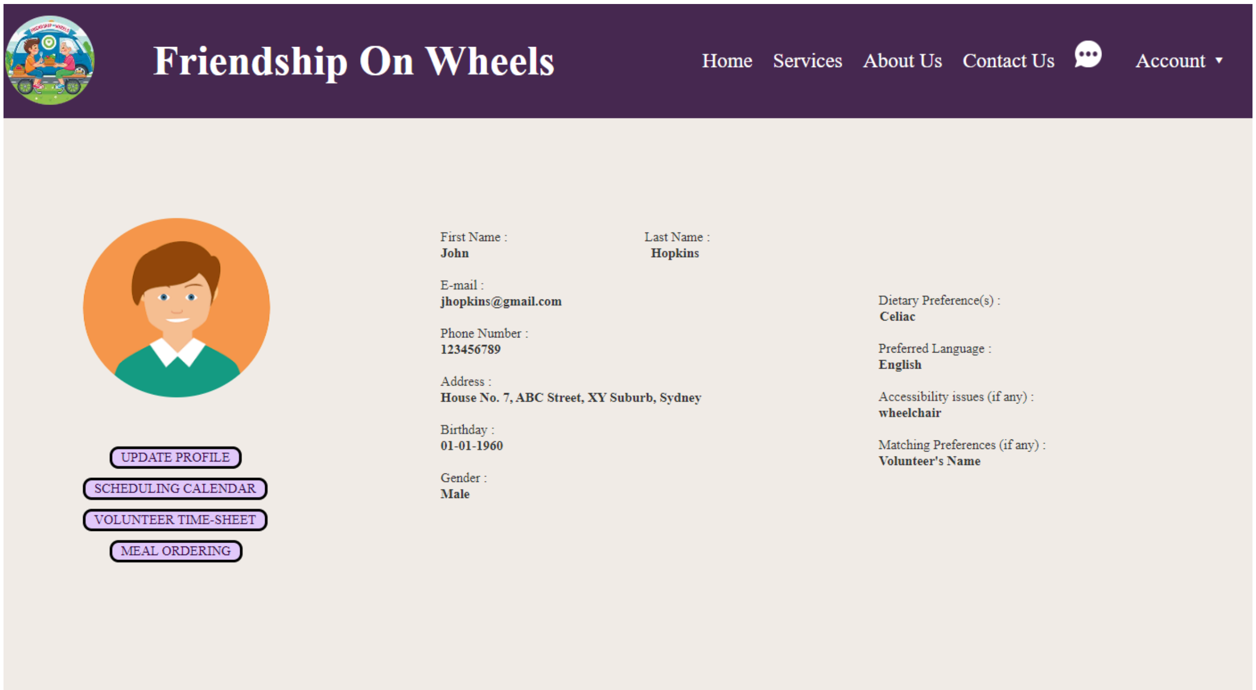
Registration Page

Our first functional requirements specifies the ability to register for both volunteers and requesters. This has been implemented in our system to be the first point of contact for users to be able to continue to use our enlisted services. At the time of registration, users can select whether they would like to volunteer or request our services. The registration form contains fields marked as mandatory to ensure that all necessary data is collected for admin purposes and as security measures. In adherence to Shneiderman’s Golden Rule of ‘Supporting internal locus of control’, users are prompted to complete these mandatory fields before submitting the form *(Poon, J. 2024)*. This design choice empowers users by providing clear guidance and control over the registration process.

Profile Page

Continuing on, users will have access to their profile pages to edit their personal details as needed and will have a seamless log in/out process with a click of a button. This follows our second and third functional requirement of seamless login process and profile update functionality, designed to prioritise usability and efficiency, also aligning with the HCI principle of minimising cognitive load and supporting user control.

Once logged in, depending on the registration as a volunteer or a requester, account holders are able to access different functionalities of the website. This will be further explored in the later sections.



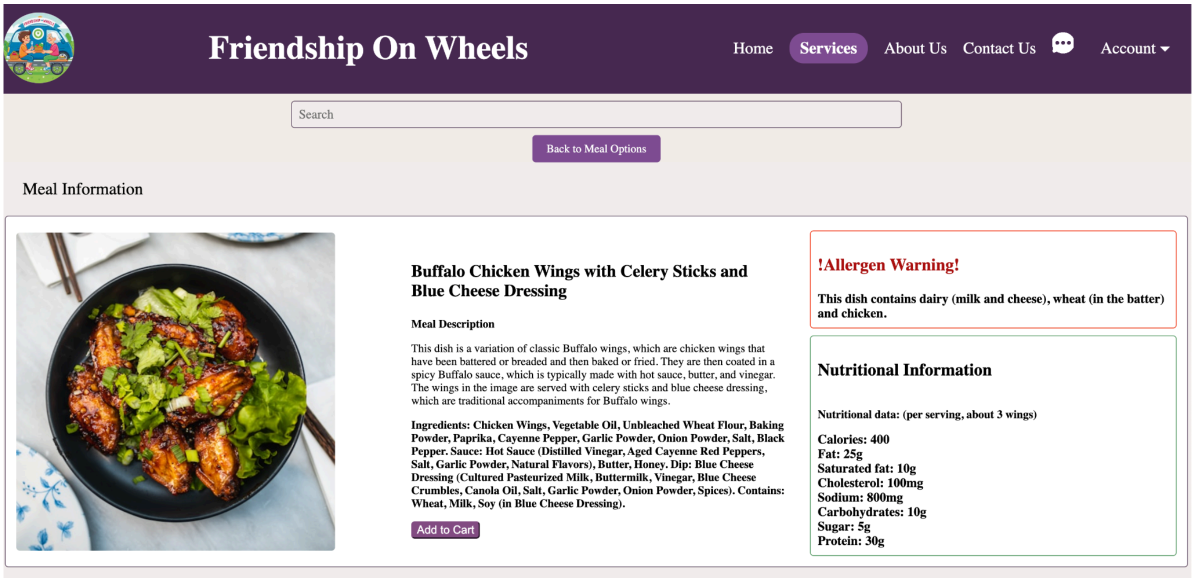
After the account creation, users are able to easily update their details in their personal detail page by editing their details and clicking the save button. For a more personalised experience, under the profile page, there are additional sections for requesters to provide further details of their dietary requirements, meal preferences and other select details. These details are not requirements for an account set up in which these informations were not collected during the registration stage. The preference entrance is solely by user choice and all details can be edited at any given time as our third functional requirement states.

Also with privacy issues large in mind, each users' personal details and any emergency contacts are kept hidden from all other users and will only be open to the assigned volunteer/requester. In addition to this, we will also be providing users the option to hide certain parts of their personal information if they are deemed non-essential.This commitment to user privacy and data protection aligns with principles of privacy and security, ensuring user trust and confidence in the system.

Following our non-functional requirement four on security, all these details, updates of any users will be strictly managed by the administrators and will only be released when necessary. That is, in instances of emergencies and critical situations only, where users' safety may be at risk.

Meal Order/Service Request

Our available meal options can be explored in the meal option page for the requesters. The page enlists a wide variety of options in a 3-column layout with images and meal names. By limiting the columns of images to 3, we ensure that all images and meal names are quickly identifiable in one glance, adhering to the HCI principle of '*Design dialogue to yield closure*.' This layout provides users with a clear and concise overview of available options, facilitating efficient decision-making as our third non-functional requirement of an easily navigable interface holds. Furthermore, detailed information for each meal is accessible under the specific meal information page, maintaining a simple and minimalistic design approach. The inclusion of clear images and the 'Add to Cart' button streamlines the ordering process, emphasising clear interaction cues for users and promoting a seamless browsing experience. This directly translates to our ninth functional requirement of users being able to place an order.



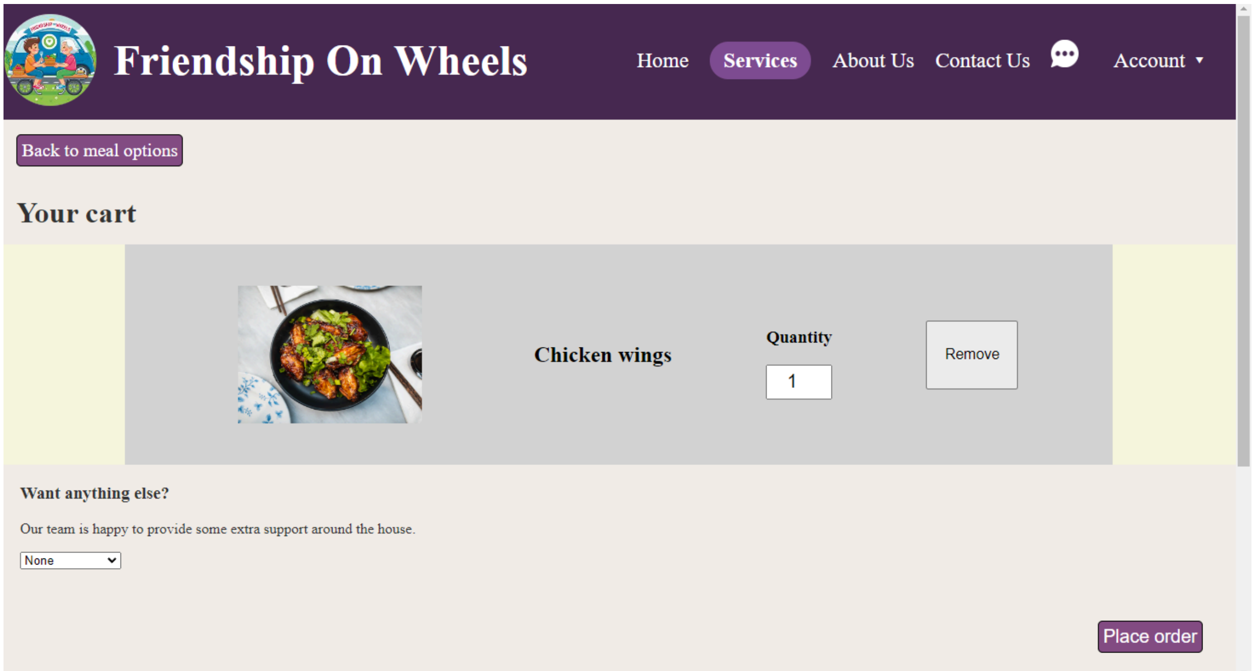
Meal descriptions are accessible by clicking any area of the box containing the meal image and meal name in our meal option page.

Ordering system

An integral component of our system, the ordering process, is precisely designed to provide users with a seamless and error-free experience following our ninth functional requirement of ordering services. The cart page serves as a central hub, displaying a comprehensive overview of the user's selections. Each item is presented clearly, with options to adjust quantities and remove items as needed, promoting user control and flexibility. Additionally, users have the opportunity to add extra services, such as Safety Checks or Friendly Meet-Ups, effortlessly integrated into the ordering interface.

To enhance usability, all essential functions and information are presented in bold text, ensuring readability, and facilitating efficient decision-making. This approach aligns with HCI principles of clarity and user-centric design, minimising cognitive load and reducing the likelihood of user error. Finally, a prominent 'Place Order' button at the bottom of the page provides users with a clear call to action, guiding them through the completion of their order with ease. Overall, our streamlined ordering process reflects our commitment to delivering a user-friendly and intuitive experience for all users.
Although not implemented on the website, our fifteenth functional requirement of intelligent meal recommendation can be incorporated into the meal ordering section where AI generated meal recommendations can suggest meals based on the users' previous order history, and their

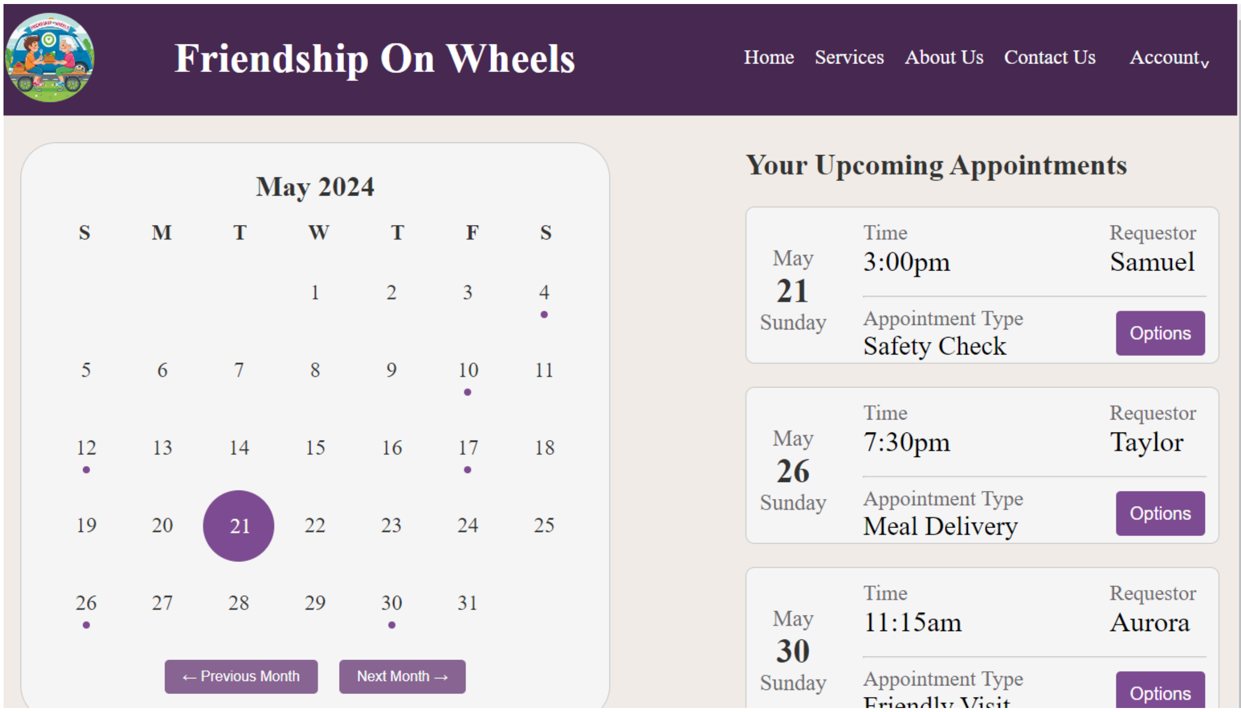
entered preferences from their profile page. This feature can be shown on the bottom section of the users' cart page, even without any previous order history, the AI will generate randomised recommendations of different cuisines.



Unique Features of Our Website

Scheduling calendar

As our seventh functional requirement states, our 'Scheduling Calendar' page provides volunteers and requesters with an easy interface to access and manage their upcoming appointments. In line with the seventh functional requirement, this feature offers a simple and intuitive way for users to comprehend and organise their service appointments for both actors and carry an additional feature of timesheet management for the volunteers. The layout features a monthly calendar on the left side, accompanied by a comprehensive list of upcoming appointments on the right. To enhance user orientation, the current date on the calendar is highlighted with a purple circle, while smaller purple dots underneath other days indicate scheduled appointments, promoting user engagement and clarity of information.



Furthermore, the schedule page incorporates HCI principles by offering interactive functionality, such as the ability to navigate to previous or next months, despite the backend implementation pending. The detailed view of upcoming appointments includes essential information such as day, time, appointment type, and requester details (although, if you are logged in as the requester, you would see volunteer details). Users are empowered with options to reschedule, cancel, or add appointments to their personal device calendar, enhancing user control and flexibility. Despite functional limitations in the buttons of the prototype, these design choices prioritise user experience and reflect our commitment to delivering a user-centric interface that aligns with HCI principles of usability and interactivity.

Timesheet for Volunteers

As previously mentioned, our timesheet feature empowers volunteers to accurately log their hours spent visiting the requester, providing both admins and the volunteers with a detailed record of their activities. This serves as a valuable tool for resolving any potential issues, such as missed visits, by offering a transparent and accountable system. In alignment with our previously mentioned seventh functional requirement, the timesheet section allows volunteers to fill in their hours, ensuring data integrity and confidentiality. Our non-functional requirement of high security and data backup (4 & 6) will be implemented in sections of not only personal details and order history but for timesheets and scheduling calendars for all users. Through this, all data and information regarding each user will be ensured to be kept private and traceable even at system failure or accidental deletion. The weekly calendar, prominently displayed at the top of the webpage and marked with green or red boxes to indicate volunteer availability, offers a clear and intuitive interface for scheduling and tracking visits. By maximising the width of the webpage, we ensure that the calendar remains easily readable and user-friendly, enhancing usability and minimising cognitive load, in line with Nielsen's heuristic of visibility of system status and flexibility and efficiency of use.

Moreover, the timesheet feature serves as a source of motivation for volunteers, providing them with a tangible record of their contributions and accomplishments in assisting requesters. This design choice aligns with HCI principles of user engagement and motivation, fostering a sense of ownership and pride in their volunteer activities. In the event of any timesheet issues, administrators are tasked with correcting discrepancies, ensuring the accuracy and reliability of the data. Overall, our timesheet feature embodies our commitment to delivering a user-centric interface that prioritises usability, accountability, and volunteer satisfaction, while also adhering to Nielsen's heuristics of error prevention and recognition rather than recall.

Live Chat Feature

According to our survey results, the majority of our audience preferred having a live chat feature, which we have integrated to facilitate deeper connections and provide immediate assistance. The live chat interface is simple and intuitive, featuring a text box for users to type messages in accordance with our last functional requirement of a 24hr live chat feature. User messages are highlighted in a blue box, while responses from our service providers are displayed in grey, ensuring clear differentiation and avoiding confusion. This design supports the HCI principle of "preventing errors" by making interactions straightforward and easy to follow and supports our third non-functional requirement of an intuitive interface for easier use.

Additionally, the live chat icon is consistently present in the header, allowing users to access this feature anytime, regardless of the webpage they are on. This adheres to the HCI principle of "visibility of system status," providing users with constant and immediate means of communication. It also aligns with the principle of "flexibility and efficiency of use," enabling quick access to support and enhancing the overall user experience. Overall, the live chat feature meets audience preferences and embodies key HCI principles to ensure it is accessible, clear, and user-friendly.

Reflection on Presentation Feedback

Upon receiving feedback on our presentation, we took deliberate steps to enhance our project components. Specifically, we improved our use case diagram as discussed earlier, ensuring it now adheres to the recommended verb-noun structure for each use case bubble. This correction addresses the identified mistakes and clarifies the diagram’s intended functions.

The remainder of the feedback we received was highly positive. As a result, we decided not to make any changes to our web design or analysis sections, to the point that we converted our presentation script into our report, with necessary additions for comprehensiveness. This approach reflects our commitment to integrating constructive feedback and maintaining high standards in our project documentation.

References

1. Babich, N. (2016) *10 Usability Heuristics for User Interface Design*, *Golden Rules of User Interface Design* . Available at: https://uxplanet.org/golden-rules-of-user-interface-design-19282aeb06b (Accessed: 07 May 2024).
2. Nielsen, J. (1994) *10 usability heuristics for user interface design*, *Nielsen Norman Group*. Available at: https://www.nngroup.com/articles/ten-usability-heuristics/ (Accessed: 08 May 2024).
3. Poon, J. 2024. User Interface. https://canvas.sydney.edu.au/courses/57239/files/36352003?module_item_id=2287230

We also wanted to reference the images we used on our website.

1. Studer Education. (2020). Community Comes Together to Fund Senior Meal Program. [online] Available at: https://www.studereducation.com/articles-and-news/community-comes-together-to-fund-senior-meal-program/ [Accessed 9 May 2024].
2. Sandy Cheu (2019). Young people considered crucial to ending ageism. [online] Australian Ageing Agenda. Available at: https://www.australianageingagenda.com.au/clinical/social-wellbeing/young-people-considered-crucial-to-ending-ageism/ [Accessed 17 May 2024].
3. Senior Care Center. (n.d.). Senior Care Center - Senior Placement Agency - Find The Best Home. [online] Available at: https://srcarecenter.com/senior-placement-agency/ [Accessed 17 May 2024]